Paraminder Dhiman

Technical Artist · Tools and Pipelines

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SUMMARY

Technical Artist with 7 years at Ubisoft, including the shipped AAA title Skull and Bones (PS5, Xbox Series X|S, PC). I build artist-facing tools and asset pipelines in MaxScript and Python across 3ds Max, Houdini and Ubisoft's in-house Anvil engine. My work also covers technical animation, cinematics and technical design, and outside the studio I edit video and run a self-hosted server platform for 20 users.

EXPERIENCE

Ubisoft India Studios — Pune, India
Technical Artist · Skull and Bones — Feb 2023 – Present
Ship building and ship customisation pipeline

- Built an ornament placement tool in 3ds Max (MaxScript) that replaced a manual Houdini workflow, cutting placement from up to 1.5 hours to about 15 minutes per ship across 22+ ships.
- Solved asset mirroring for an engine that does not accept negative scale by implementing quaternion-based mirroring, so ornaments mirror correctly across the hull.
- Rolled the placement tool into the studio pipeline, where it is used by art teams in local and overseas studios.
- Took over a broken, inherited sail animation pipeline (SSDR cloth-to-bone baking in Houdini), reverse-engineered how it works and became its sole owner on the project; now repairing it.
- Built a new ship spawning system on the engine's entity-building tool to assemble and spawn ship meshes.
- Optimised Houdini ship-building workflows to speed up artist iteration.
- Profiled and optimised shaders and materials to keep ship content within performance budgets.
- Mentored junior technical artists in MaxScript and Python scripting, Houdini and Anvil workflows.

Ubisoft India Studios — Mumbai, India
Junior Technical Artist · Skull and Bones — Oct 2021 – Feb 2023
Pets (player vanity items)

- Delivered 50+ pets, supporting each one from concept art through ready-to-animate (RTA) setup to final integration in the engine.
- Rebuilt the pet integration pipeline and the pet spawning system.
- Restructured and rebuilt the full pet animation graph in the in-house animation graph tool.
- Worked across 3ds Max, Blender, Maya and the Anvil engine.

Ubisoft India Studios — Mumbai, India
Junior Technical Level Designer (Cinematics) · Prince of Persia: The Sands of Time Remake — Nov 2019 – Oct 2021
Original Ubisoft Mumbai and Pune version (unreleased)

- Integrated and validated character animations for cutscenes in the in-house timeline-based cinematics editor, working with character, audio and lighting tracks.
- Lit around 20 cutscenes.
- Owned and resolved all cinematics-related bugs.
- Prepared animation data in MotionBuilder for cutscenes, with cleanup where needed.
- Scripted cinematic triggers.
- Edited cinematic video in Adobe Premiere Pro.
- Maintained cutscene documentation for the team.

PROJECTS

Black Hole · Blender and Unreal 2023 – 2026

Personal · Blender, Python, OSL · Unreal Engine 5, HLSL

- **Black Hole Studio** is a Blender extension that renders a black hole by bending light the way general relativity does. Fast mode bakes every light path into a lookup image so it runs on the GPU and in Eevee; Exact mode traces each ray in an OSL shader in Cycles. I designed it and wrote much of the code, with parts developed with AI assistance. Free on Blender Extensions; code on GitHub.
- **Black Hole Shader** is a real-time Unreal Engine 5 material with a custom HLSL node: light bent by the Schwarzschild equations, a volumetric accretion disk, Doppler beaming and blackbody colour. Version 2.0 made the bending physically accurate, with artist controls on top. Free on Fab.

Blender Extensions 2026

Personal · Blender, Geometry Nodes, Python

- **Unravel** tears cloth: paint where it should tear, and it cuts the holes, thickens their edges and grows frayed threads, built in Geometry Nodes with a Python add-on. Published on Blender Extensions.
- **UV Island Colorizer** gives every UV island its own colour and bakes a colour ID map, across many meshes at once. Published on Blender Extensions.
- **Positive Mirror** mirrors objects with a positive scale instead of −1, so normals and exports stay correct. Code on GitHub.

Home Server and Automation 2026

Personal · Unraid, Docker, Python

- Designed and run a self-hosted services platform for 20 users, with Docker containers, secure remote access (Tailscale, Cloudflare Tunnels), network-wide DNS and DHCP, and scheduled backups.
- Wrote a Python Discord bot (discord.py) that automates user onboarding and account provisioning, deployed as a container and versioned in Git.

More projects

Every project, with how and why each was built, is on my portfolio at paraminder-dhiman.github.io.

SKILLS

Scripting and programming
MaxScript, Python, C#, HLSL, C++ (working knowledge), VEX (working knowledge)

Technical animation
Animation graphs, Cloth-to-bone baking (SSDR), Animation data preparation in MotionBuilder

DCC software
3ds Max, Houdini, Blender, Maya, MotionBuilder

Video
DaVinci Resolve (editing, Fusion, colour grading), Adobe Premiere Pro

Technical art
Tool development, Asset integration pipelines (RTA), Shader and performance profiling

Cinematics and technical design
Timeline-based cutscene assembly, Cutscene lighting, Cinematic trigger scripting

Engines and pipeline
Anvil (Ubisoft in-house), Unreal Engine 5, Perforce, Git

Infrastructure
Docker, Unraid, Home networking and remote access

EDUCATION

Abertay University Professional Masters in Games Development (MProf)
Dundee, UK • 2019

Supinfogame Rubika Bachelor cycle in Video Game, Game Design specialisation (3 years, 180 ECTS)
Pune, India • 2015 – 2018